



Ensaie 3A/MS/Masters P7 & SFA 2024-2025.

Exam “Copulas and Financial Applications”

2 hours. Any document & calculator is allowed.

Exercise.

Consider a Marshall-Olkin copula defined by

$$C(u, v) = u^{1-\alpha}v\mathbf{1}(u^\alpha \geq v^\beta) + uv^{1-\beta}\mathbf{1}(u^\alpha < v^\beta), \quad u, v \in [0, 1],$$

for some parameters $0 < \alpha, \beta < 1$.

1. Check that the margins of C are uniform.
2. Calculate f_C , the density of C with respect to the Lebesgue measure, on the domain $A_1 = \{(u, v) \in (0, 1)^2 \mid u^\alpha > v^\beta\}$. Similarly, calculate f_C on the domain $A_2 = \{(u, v) \in (0, 1)^2 \mid u^\alpha < v^\beta\}$.
3. Calculate $\int_{A_1} f_C(u, v) du dv$, $\int_{A_2} f_C(u, v) du dv$, and their sum. Conclude.
4. Show that

$$\int_0^1 \int_0^1 C(u, v) du dv = \frac{1}{2} \left(\frac{\alpha + \beta}{2\alpha - \alpha\beta + 2\beta} \right).$$

Deduce that the Spearman’s rho of C is

$$\rho_C = \frac{3\alpha\beta}{2\alpha - \alpha\beta + 2\beta}.$$

5. We observe a sample $(X_i, Y_i)_{i=1, \dots, n}$ of realizations from a random vector (X, Y) , whose copule is C . When $\alpha = \beta$, propose an estimator of α .

Problem.

Consider a strictly decreasing convex function $\phi : (0, 1] \mapsto [0, +\infty)$ verifying $\phi(1) = 0$, $\lim_{t \rightarrow 0} \phi(t) = +\infty$. Moreover, let a convex function $A : [0, 1] \mapsto [1/2, 1]$ such that $\max(t, 1-t) \leq A(t) \leq 1$ for all $0 \leq t \leq 1$. The maps A and ϕ have a continuous first derivative on $[0, 1]$ and $(0, 1]$, respectively.

A bivariate function $C_{\phi,A} : [0, 1]^2 \rightarrow [0, 1]$ is said to be Archimax if and only if it can be written in the form

$$C_{\phi,A}(u, v) = \phi^{-1}\left(\left\{\phi(u) + \phi(v)\right\}A\left(\frac{\phi(u)}{\phi(u) + \phi(v)}\right)\right),$$

for all $0 \leq u, v \leq 1$.

Part 1.

1. Show that $C_{\phi,A}(u, 1) = u$, $C_{\phi,A}(1, v) = v$, $C_{\phi,A}(u, 0) = 0$ and $C_{\phi,A}(0, v) = 0$ for all (u, v) in $[0, 1]$.

2. Show that $C_{h\phi,A} = C_{\phi,A}$ for any positive real number h .

3. Show that the Archimedean copulas and the Extreme value copulas are Archimax functions.

Denote $\partial_1 C_{\phi,A}(u, v)$ the value at (u, v) of the partial derivative of $C_{\phi,A}$ with respect to its first component.

4. Prove that

$$\partial_1 C_{\phi,A}(u, v) = \frac{\phi'(u)}{\phi'(C_{\phi,A}(u, v))} \left\{ A\left(\frac{\phi(u)}{\phi(u) + \phi(v)}\right) + A'\left(\frac{\phi(u)}{\phi(u) + \phi(v)}\right) \frac{\phi(v)}{\phi(u) + \phi(v)} \right\}.$$

5. For any numbers a, b, c, d in $[0, 1]$, denote $V_{(a,b] \times (c,d]}(C_{\phi,A})$ the $C_{\phi,A}$ -volume associated to the “box” $(a, b] \times (c, d]$. Prove that

$$V_{(a,b] \times (c,d]}(C_{\phi,A}) = \int_a^b \left\{ \partial_1 C_{\phi,A}(u, d) - \partial_1 C_{\phi,A}(u, c) \right\} du.$$

6. Define the map $\beta : t \mapsto A(t) + A'(t)(1-t)$, $t \in [0, 1]$. For any (t_1, t_2) in $[0, 1]$, $t_1 \leq t_2$, prove that

$$\beta(t_2) - \beta(t_1) = \int_{t_1}^{t_2} \left\{ A'(\xi) - A'(t_1) \right\} d\xi + (A'(t_2) - A'(t_1))(1-t_2).$$

Deduce that β is increasing and that $v \mapsto \partial_1 C_{\phi,A}(u, v)$ is increasing, for any $u \in [0, 1]$.

7. Show that the map $C_{\phi,A}$ is a bivariate copula.

Part 2.

Consider X and Y two survival times (positive random variables), whose survival functions are denoted $\bar{F}$ and $\bar{G}$ respectively. Moreover, consider another positive random variable Z whose cumulative distribution function is M . Conditionally on $Z = z$, the joint survival function of (X, Y) is $(S(x)T(y))^z$, for two “baseline” survival functions $S : \mathbb{R}^+ \rightarrow [0, 1]$ and $T : \mathbb{R}^+ \rightarrow [0, 1]$.

8. Show that the joint survival function of (X, Y) is

$$P(X > x, Y > y) = \int_0^{+\infty} (S(x)T(y))^z M(dz), \quad x \geq 0, y \geq 0.$$

9. Denote by $\psi : \mathbb{R}^+ \rightarrow [0, 1]$ the Laplace transform of Z . Prove that

$$\psi^{-1}(\bar{F}(x)) = -\ln S(x), \quad x \geq 0, \text{ and}$$

$$\psi^{-1}(\bar{G}(y)) = -\ln T(y), \quad y \geq 0.$$

10. Prove that the survival copula of (X, Y) is Archimedean, and that ψ^{-1} is its generator.

Part 3.

Consider X and Y two survival times (positive random variables), whose survival functions are denoted $\bar{F}$ and $\bar{G}$ respectively. Moreover, consider another positive random variable Z whose cumulative distribution function is M . Conditionally on $Z = z$, the joint survival function of (X, Y) is $C_A(S^z(x), T^z(y))$, for two “baseline” survival functions $S : \mathbb{R}^+ \rightarrow [0, 1]$ and $T : \mathbb{R}^+ \rightarrow [0, 1]$ and the bivariate copula C_A , where

$$C_A(u, v) = \exp\left(\ln(uv)A\left(\frac{\ln u}{\ln(uv)}\right)\right), \quad u, v \in (0, 1].$$

Here, the map $A : [0, 1] \rightarrow [0, 1]$ is defined as in Part 1.

11. Write the joint survival function of (X, Y) .

12. Denote by $\psi : \mathbb{R}^+ \rightarrow [0, 1]$ the Laplace transform of Z . Prove that

$$\psi^{-1}(\bar{F}(x)) = -\ln S(x), \quad x \geq 0, \text{ and}$$

$$\psi^{-1}(\bar{G}(y)) = -\ln T(y), \quad y \geq 0.$$

13. Prove that the survival copula of (X, Y) is Archimax.

14. Assume that ψ is known and that the map A belongs to a parametric family, i.e. it is denoted A_θ , $\theta \in \Theta \subset \mathbb{R}^p$. We observe a sample $(X_i, Y_i)_{i=1, \dots, n}$ of independent realizations of the bivariate random vector (X, Y) . Propose an inference strategy to estimate the true parameter θ .